

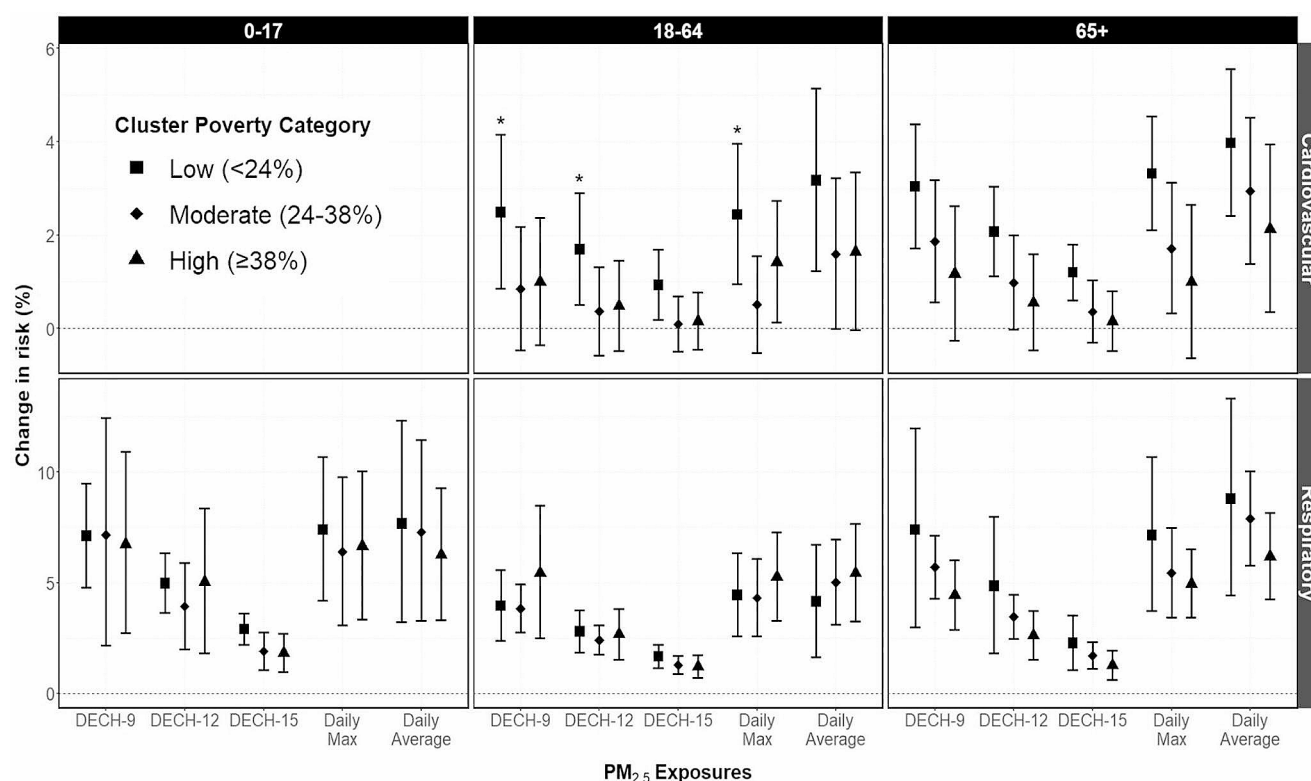


Fig. 4 Excess risk and confidence intervals for the association between PM_{2.5} exposure metrics and EDVs per each IQR increase of PM_{2.5} exposure metrics, stratified by patient age and cluster-level poverty. Asterisk (*) represents statistically significant difference compared to

the reference group, moderate poverty clusters. For cardiovascular-related outcomes, results of a 7-day lag were presented; for respiratory outcomes, results of a 3-day lag were presented

living in low poverty clusters showed higher excess risk estimates than those living in the moderate and high poverty strata but with wide CIs. As for children, excess risks based on DECH-15 suggested slightly higher risks of respiratory EDVs in the low poverty clusters, similar to the trend of daily average, but no clear variation was observed on daily max, DECH-9 or DECH-12 exposures. For adults (18–64 years), high poverty clusters showed slightly higher estimates of respiratory risks than the other two strata, as observed on DECH-9 and daily maximum, though these differences were not statistically significant.

These findings suggest the possible effect measure modification by poverty level on the relationship between PM_{2.5} and cardiovascular-related outcomes for both adult groups and respiratory outcomes for children and older adults. Results based on subdaily metrics of PM_{2.5} generally reflected the trend observed on daily average concentration, although the magnitude and direction of such modification differed by metric. Of the subdaily metrics, DECH-9 and daily maximum PM_{2.5} produced estimates most similar to daily average PM_{2.5}, whereas DECH-12 and DECH-15 were more attenuated, mirroring the results of the primary model.

Sensitivity analyses

As a sensitivity test, co-pollutants, O₃ and NO₂, were adjusted in the model (Fig. 5). PM_{2.5} metrics were robust in the co-pollutant model with O₃ for both cardiovascular- and respiratory-related EDVs. When adjusting for NO₂, the associations with cardiovascular-related EDVs remained stable. DECH-12 of PM_{2.5} showed similar associations with cardiovascular-related outcomes when adjusted for DECH-5 and DECH-13 of NO₂, compared to no adjustment. However, NO₂ appeared to attenuate the association between PM_{2.5} and respiratory EDVs across all PM_{2.5} metrics.

Discussion

We evaluated subdaily exposure to PM_{2.5} and found that subdaily metrics of PM_{2.5}, daily maximum and DECH-9, generally showed similar elevated risks and patterns as daily average PM_{2.5}, the current PM_{2.5} standard metric, on cardiorespiratory EDVs throughout our study period of 2018 to 2020 in Southern California. Subdaily metrics with higher thresholds (DECH-12 and –15) were comparatively attenuated. Patients in the older adult group consistently had

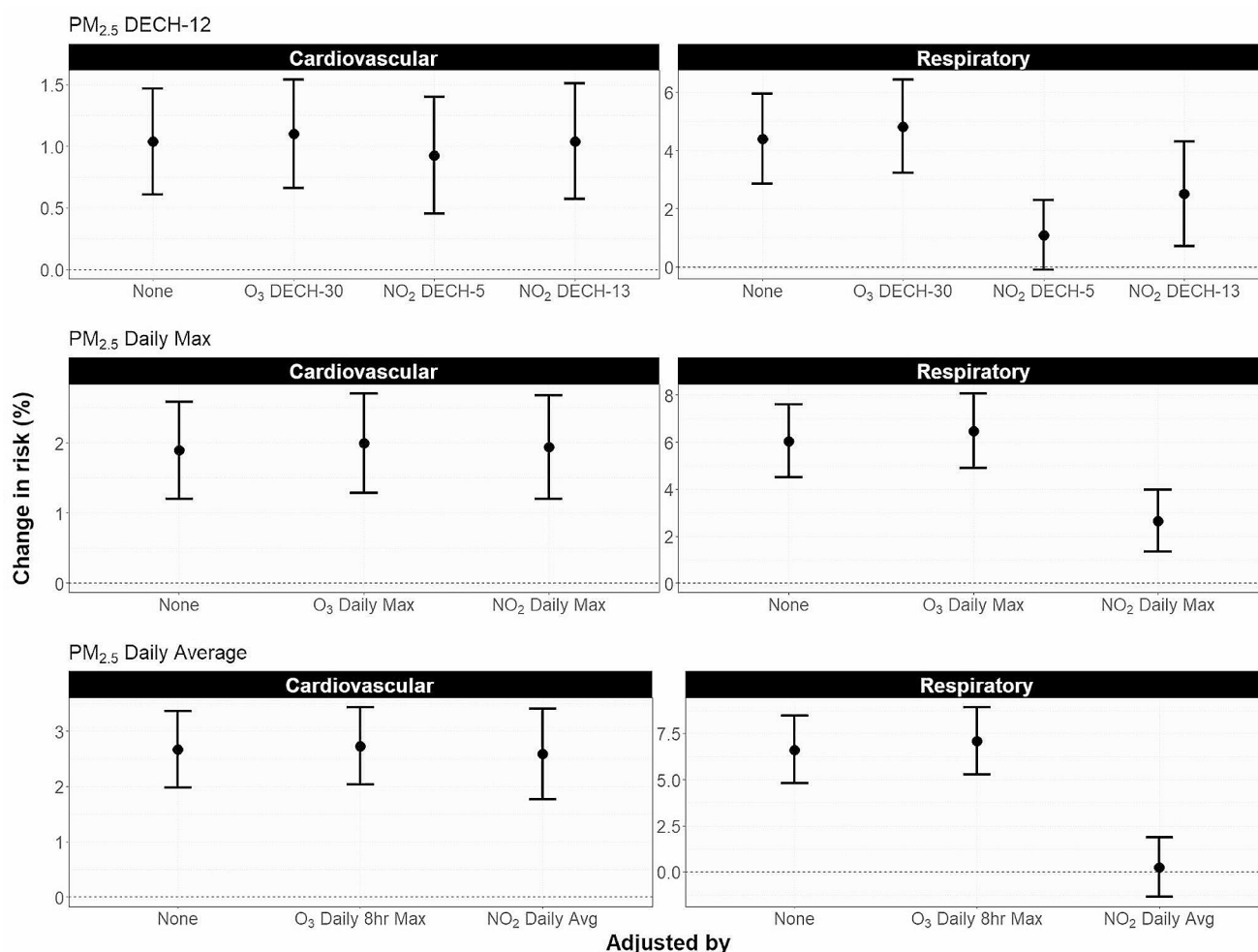


Fig. 5 Excess risks and confidence intervals for the association between $PM_{2.5}$ exposure metrics and EDVs per each IQR increase of exposure metrics, with and without co-pollutant adjustment. For cardiovascular-

related outcomes, results of a 7-day lag were presented; for respiratory outcomes, results of a 3-day lag were presented

a higher excess risk of cardiovascular-related outcomes, and the children and older adult groups showed higher excess risks of respiratory outcomes compared to adults. Race/ethnicity and neighborhood poverty level did not show consistent modification effect. Associations between $PM_{2.5}$ and cardiorespiratory outcomes were robust to the adjustment with O_3 and NO_2 , except that NO_2 attenuated the association of $PM_{2.5}$ and respiratory outcomes, indicating that traffic emissions may account for the majority of adverse respiratory outcomes.

Comparison to other studies

Our findings are in agreement with earlier research conducted outside of California, specifically, the associations of cardiorespiratory outcomes with subdaily $PM_{2.5}$ exposure were comparable or smaller than those with the daily average metric. Several studies in China have examined

the associations between DECH metrics and various cardiorespiratory outcomes, such as hospitalization of chronic obstructive pulmonary disease, outpatient visits of hypertension, and all-cause EDVs, reporting a similar reduction in effect when using DECH (Jiao et al. 2020; Lin et al. 2021; Zhang et al. 2021). However, another study looking into outpatient visits of acute lower respiratory infections among children reported stronger associations with DECH metrics than daily average $PM_{2.5}$ (Xiao et al. 2022). Such a difference could be due to the differences in study populations and medical systems, type and severity of health outcomes, and/or $PM_{2.5}$ sources and components. Therefore, results of studies in China may not be generalizable to California, which is one of the motivations of this study. One thing to note is that our data had many zero values for DECH-12 and -15, namely, the non-zero values of these DECH metrics only covered exposure of a portion of time period; whereas in most of the Chinese studies, DECH metrics had

non-zero values for almost the entire study periods, so the DECH results may be more close to the daily metric, such as DECH-9 in our study.

A number of studies using other subdaily metrics, such as daily maximum, hourly concentration, or exposure averaged across several hours, also obtained similar results. For example, the review by Buteau et al. (2019) found that the pooled odds ratios (ORs) of hospitalization for asthma and bronchiolitis in children < 5 years were 1.01 (95% CI: 0.98, 1.03) and 1.02 (0.93, 1.10) per 10 $\mu\text{g}/\text{m}^3$ increment in the hourly maximum concentration and daily mean of $\text{PM}_{2.5}$. A study in Israel using telemedicine data obtained similar ORs, 1.008 (95% CI: 0.998, 1.018) and 1.006 (0.995, 1.018), respectively, for any cardiac event associated with a 10 $\mu\text{g}/\text{m}^3$ increase in 6-h and 24-h average $\text{PM}_{2.5}$ (Nirel et al. 2018).

In this study, we examined the exposure up to a 14-day lag. Previous studies usually examined shorter lags and found that cardiovascular effects tended to be most influential on the same day or immediate lag periods (up to two days) (USEPA 2019); respiratory effects appeared more transient, with largest effect size observed within the hour of exposure and lagged up to 48 h. There was very limited evidence on associations for longer lags, i.e., 7-day or 14-day lags. Kim et al. (2012) observed strongest cardiovascular effects on the day 0 and another bump of risks around the lag 7th day, then the effect diminished thereafter. Along with this evidence, we observed cumulative exposures over longer lags were associated with greater risk increments in cardiovascular-related outcomes across all metrics. Respiratory risks associated with daily average $\text{PM}_{2.5}$ also accumulated along lags, peaking at lag 7 days. Such trends were also observed in earlier California studies (Bi et al. 2020; Delfino et al. 2014). We further found that, the respiratory risks associated with subdaily metrics remained after lag 7 days, compared to the decreasing trend of daily average $\text{PM}_{2.5}$, implying prolonged respiratory-related effects that were not reflected by daily average metric.

Health disparity is another focus of this study. Previous work in the U.S. have suggested cardiorespiratory health disparities with varying evidence that Black and Hispanic groups have higher exposure to air pollution and/or increased cardiorespiratory risks (Bell and Ebisu 2012; Boing et al. 2022; Liu et al. 2021), although some found higher $\text{PM}_{2.5}$ -associated respiratory hospitalization rate among White individuals compared to the overall population (Mehta et al. 2021). We observed higher excess risks of respiratory EDVs and lower excess risks of cardiovascular-related EDVs among Hispanic population associated with both subdaily and daily $\text{PM}_{2.5}$ metrics, compared to non-Hispanic White, likely driven by the age difference between the two groups. With a younger average age, the Hispanic

group contains a higher proportion of children vulnerable to respiratory outcomes. Furthermore, differing from previous studies, we observed more cardiovascular-related EDVs among people living in higher SES areas, for both adult groups. This might be attributed to multiple reasons: better health insurance coverage for people of higher SES; language and transportation barriers to access healthcare among lower SES groups; or lack of sick leave to seek medical care especially for the low-income workers (Peipins et al. 2012).

There are several established pathophysiological pathways that would explain the positive associations between $\text{PM}_{2.5}$ and the cardiorespiratory outcomes that we observed. Inhaled $\text{PM}_{2.5}$ can penetrate to the deep lung and enter the bloodstream, causing not only respiratory tract inflammation but also systematic inflammation responses and oxidative stress that may further trigger asthma and alter hemostasis which may also lead to MI, and heart failure, increase heart rate variability, coronary heart diseases, and increased blood pressure (USEPA 2019). Pulmonary inflammation and decreases in lung functions were largely observed as well, although these effects are considered transitory (USEPA 2019).

Implications of utilizing subdaily $\text{PM}_{2.5}$ metrics

The observed elevated risks for cardiorespiratory outcomes with exposure to $\text{PM}_{2.5}$ DECH-9 lend support for the new U.S. national ambient annual $\text{PM}_{2.5}$ standard values and adds to the body of evidence that indicates significant health effects below 12 $\mu\text{g}/\text{m}^3$. $\text{PM}_{2.5}$ exposure above 9 $\mu\text{g}/\text{m}^3$ shows similar risks of cardiovascular- and respiratory-related EDVs associated with the daily average metric.

In our findings, both daily maximum and DECH metrics readily follow the daily average metric, which aligned with USEPA's early determination. Daily average metric appeared to cover most of the cardiorespiratory-related effects suggested by the daily maximum and DECH metrics. Elevated subdaily transient exposures like fires and fireworks were not frequent enough to influence the overall associations in our study area, but the particular $\text{PM}_{2.5}$ components captured by DECH-12 and -15 may be more toxic than those in the regional background or more toxic above certain threshold, showing longer lagged respiratory effects. In this manner, subdaily measurements of $\text{PM}_{2.5}$ would be relevant in historically highly polluted areas and in the event of air pollution accidents. However, given the confounding effects by NO_2 , such prolonged effects need to be further confirmed.

Based on the existing results, the daily average metric was sufficient to characterize ambient $\text{PM}_{2.5}$ exposure and associated cardiorespiratory health outcomes, and health

guidance based on daily metric could provide sensible protection to the public in general. The hourly Air Quality Index (AQI) appears to be an easy and efficient way to communicate with the public, as hourly AQI is calculated by referencing hourly $PM_{2.5}$ and annual/daily National Ambient Air Quality Standards. Nevertheless, given the inconclusive finding, we recommend that both sensitive populations as well as healthcare providers to stay vigilant for elevated subdaily exposure and prepare for potential cardiorespiratory effects in their localities, particularly if they live in or provide care in polluted areas. Future studies in other densely populated areas, allowing for more socioeconomic stratifications and investigation of specific diseases, are needed to verify our findings, whereas conducting similar analysis in less populated areas would be challenging.

Strengths and limitations

Our study has many strengths. We utilized relatively complete, hourly gridded exposure data for a large portion of Southern California to create subdaily metrics. We were able to control for co-pollutants to parse out the individual effect of $PM_{2.5}$ on cardiorespiratory outcomes. Despite the challenge of low EDV counts at ZCTA level, we were able to leverage a spatial clustering procedure to create larger geographic units of observations based on sociodemographic likeness and exposure variability to increase outcome counts, especially in rural areas. This regionalization method may be a valuable tool for future analyses involving sparse counts and spatial autocorrelation of observations, making it possible to investigate potential effect measure modification by sociodemographic factors.

On the other hand, our analyses have several limitations. First, the EDV records came with date but no exact admission time stamp, which limited the scope of analysis to the daily level and the subdaily metrics we can use. Thus, we were not able to identify which particular hour(s) of exposure were most relevant. Second, the EDV counts at each geographic unit were small, even after aggregating to the cluster level, with many zeros in sparsely populated areas. These factors restricted our ability to look at specific health outcomes and stratification analysis by potential confounders. Several Chinese studies in recent years have reported specific types of diseases, such as ST-segment-elevation MI and acute lower respiratory infections, associated with transient exposure at hourly scale (Chen et al. 2022; Jiang et al. 2023; Lei et al. 2022). However, we have neither the health data specified to hourly resolution nor enough cases to look at specific types of diseases. Considering that the Los Angeles metropolitan area is one of most populated areas in U.S., future studies may limit their study populations to urban areas or target more populated areas, e.g., New York City, to

achieve higher case counts, but the demographics of these areas may not be representative of the U.S. general population. Third, there was likely exposure misclassification due to the regionalization process. While we attempted to maximize $PM_{2.5}$ similarity within-clusters, we lost geographic variation to some degree. This issue is common when aggregating information across geographic units of varying sizes and arbitrary administrative borders (Parenteau and Sawada 2011). However, the exposure misclassification was likely non-differential (random and not systematic) in our dataset, and therefore our estimates would be biased toward the null.

Conclusions

Our findings confirmed that subdaily exposures to elevated ambient air pollution were associated with increased adverse cardiorespiratory outcomes that were basically captured by the daily average metric. Peak subdaily exposure may have respiratory effects over an extended period, longer than the daily average metric implied, but the effect estimate was still within the quantification of the daily average metric. Further studies to consider prolonged risks and specific health outcomes are warranted, though having sufficient case counts for fine spatial resolution may be challenging.

With both subdaily and daily metrics, we observed elevated excess cardiovascular risks among older adults and elevated excess respiratory risks among both older adults and children compared to adults aged 18–64 years. We also observed excess risks of cardiorespiratory EDVs associated with ambient $PM_{2.5}$ exposure among low-poverty neighborhoods. More research is needed to investigate these associations and uncover the underlying reasons.

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Ethics approval The use of emergency department visit data was approved by the Institutional Review Board of the State of California Committee for the Protection of Human Subjects.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11869-024-01583-6>.

Acknowledgements We thank Ms. Dharshani Pearson and Mr. Daniel Sultana (California Office of Environmental Health Hazard Assessment) for their insightful comments on the manuscript.